

MathGov and RippleLogic Foundations Primer v3.4

Version 3.4 (Informative)

Version 3.4, synchronized companion edition for Canon v11.4 / SGP v6.4 / ripple.md v4.4 / Agent System v11.4 / Primer v3.4 / Aligners Sheet v4.4. Package line: MathGov Core Release 2026.09 applies the Complete Ready Release, CSV Upgrade, and Falsification Discipline Patch, carrying forward the Term Discrimination and Semantic Stability discipline, including discriminator, dependency-position, do-not-collapse, and term-refusal discipline.

Companion to Canon v11.4 / SGP v6.4 / ripple.md v4.4 / Agent System v11.4 / Primer v3.4 / Aligners Sheet v4.4

MathGov Institute for Ethical Systems Design

Release alignment (v11.4)

This companion document is synchronized with MathGov Core Release 2026.09 v11.4. The public cascade is RG -> RF -> TRC -> CSV -> RLS; the formal shorthand is RG/RSG -> RF/NCRC -> TRC -> CSV -> RLS. RF is the public Rights Floor layer and NCRC is its formal non-compensatory predicate. CSV means Containment and Structural Viability. Detailed cascade definitions are maintained in the Canon, the RippleLogic Cascade Standard, and the CSV Gate Standard; this file applies those invariants without repeating the full release note.

Abstract

This primer is the human doorway into MathGov, RippleLogic, and the Sentience Gradient Protocol. It explains why the framework exists, what problem it addresses, what it claims, what it does not claim, how the main pieces fit together, and why material terms must be discriminated before they can govern decisions. It is informative only: the Canon and companion specifications remain the source of normative force.

Plain-language summary

The Rights Floor (RF) is the public-facing name for the Non-Compensatory Rights Constraint (NCRC). In plain terms, the Rights Floor means that protected rights violations cannot be averaged away by downstream benefits, majority preference, or an attractive RippleLogic Score.

Decisions ripple outward. They affect people, institutions, ecosystems, future generations, and sometimes intelligences that are not yet well understood. The MathGov framework, using RippleLogic, gives readers a structured five-level method: Reality Grounding, Rights Floor (RF), TRC, CSV, and RLS. In simpler words: ground the facts, protect the Rights Floor, bound ruin, contain and verify the whole, then choose the best ripple. The framework makes ethical reasoning auditable rather than merely rhetorical. v3.4 integrates the Complete Ready

release discipline while preserving a simple public discipline: ground the words before trusting the score.

How to read this primer

Begin with Sections 1 and 2 for the problem and basic identity of the framework. Then read Sections 3, 3A, 3B, 3C, and 22 for layer discipline, primitive audit, reference-structure reasoning, Category Grounding, and Term Discrimination. Use later sections to understand the cascade, SGP, audit posture, and release boundaries.

Section 1. The Problem MathGov Is Built To Address

Human authority alone is not stable enough for civilization-scale decision making. Humans can be wise, compassionate, courageous, and visionary. Humans can also be distorted by fear, greed, coercion, propaganda, concentrated power, institutional drift, tribal loyalty, short-term incentives, and survival pressure.

That is the surface problem. The deeper problem is structural: complex systems frequently reward local optimization while concealing larger-system harm.

A company can improve quarterly results while burning out its people. A nation can grow GDP while degrading the ecosystems supporting its survival. A government can claim stability while suppressing rights. An AI lab can increase capability while making alignment harder to reason about. From inside each system, the move can look rational. Zoom out far enough, and the pattern becomes dangerous.

Local optimization. Global degradation.

That is one of the central fractures MathGov and RippleLogic are designed to address. The goal is not to replace human judgment with a machine. The goal is to make judgment more transparent, auditable, less corruptible, less arbitrary, less captured by power, and more accountable to the full field of consequences.

Section 2. What MathGov Is, and Is Not

2.1 What MathGov is

MathGov is a reality-seeking decision architecture. RippleLogic is its decision engine. SGP is its moral status and protection protocol. Together they constitute a proposed ethical operating-system architecture for governance, AI alignment, institutional design, and high-stakes decision-making under complexity.

2.2 What MathGov is not

MathGov is not a claim of final truth. It is not a substitute for human judgment, lawful authority, or democratic legitimacy. It does not claim that all values reduce perfectly to

numbers. It does not claim that current AI systems are conscious. It does not claim to have solved alignment. It does not claim universal consent across cultures.

What MathGov claims is more modest and more useful: that ethical and governance decisions become harder to corrupt, harder to hide, harder to capture, and easier to test against reality when their primitives are declared, their assumptions are exposed, their weights are visible, their rights floors are protected, their uncertainty is modeled, and their reasoning is auditable.

Architecture-final is not ProofPack-final. A framework can be coherent, specified, and useful before it is empirically validated, schema-complete, validator-complete, or Tier-4-ready; those stronger statuses require separate evidence and replay materials.

2.3 Layer discipline

MathGov becomes clearer when each kind of statement is kept in its proper layer. It does not treat ontology, ethics, mathematics, evidence, governance, and interpretation as the same thing.

Operational ontology. Union-Based Reality describes the decision-relevant structure of interdependence: beings and systems exist inside relationships, dependencies, and consequences. This is an operational claim for governance-relevant domains, not a final metaphysical proof of all being.

Reality, decision-relevant reality, reality surface, evidence, and interpretation are related but not identical. Reality is the evaluator-independent structure of constraint and consequence that holds whether or not any agent perceives, admits, models, or has power over it. Decision-relevant reality is the portion of that structure that materially bears on the decision. A reality surface is the portion actually captured, represented, or modeled in the run. Truth, for a decision claim, means correspondence to decision-relevant reality as supported by the declared reality surface and evidence trace. Evidence is the trace by which a claim is tested against the relevant surface. Interpretation is the judgment that connects evidence, uncertainty, values, and action. MathGov does not claim perfect access to reality; it requires declared, auditable, challengeable contact with the relevant reality surface.

Root reality doctrine. MathGov defines reality, for decision-making, as the evaluator-independent field of constrained structures that, under transition, produce consequences. Structure is the stable arrangement of relevant differences. Constraint is what limits and enables possible transition. Consequence is what follows when an action, event, policy, or decision enters that constrained structure. In public language: Reality is what exists, what constrains, and what follows.

Normative commitment. Union-Based Ethics states what the framework is committed to protecting and optimizing: rights floors, reduced suffering and domination, catastrophic-risk limits, containment, and flourishing across affected stakeholders. These commitments are explicit rather than hidden inside equations.

Ethical-reality classes. MathGov distinguishes measured or indicator-supported ethical reality, such as injury, mortality, pollution, access, deprivation, or ecological exposure; inferred or uncertainty-bound ethical reality, such as sentience, suffering, agency restriction, psychological harm, or future risk; and explicit normative commitment, such as rights floors, dignity, non-domination, and future-generation standing. Each class requires a different evidence and audit posture.

Formal decision machinery. RippleLogic converts those commitments into gates, scores, weights, audit fields, and decision states. Formal machinery helps make judgment inspectable, but formal consistency alone is not proof that a decision is right.

Evidence and measurement. Impacts, probabilities, sentience estimates, stakeholder maps, and weights require evidence. Where evidence is weak, the framework must disclose uncertainty, run sensitivity checks, escalate, or refuse the stronger claim.

Governance and interpretation. PCCs, public rationales, release manifests, and educational explanations make decisions accountable. They do not create truth by themselves. They make claims easier to inspect, contest, and repair.

Plain-language rule. Do not let one layer pretend to be another. Compliance is not correctness. Plausibility is not evidence. Intelligence is not sentience. Protection is not authority. Scoring is not admissibility. Refusal is not failure; it is a guard against false certainty.

Five-level method. RippleLogic is now best taught inside the MathGov framework as: Reality Grounding -> Rights Floor -> TRC -> CSV -> RLS. Formal shorthand: RG/RSG -> RF/NCRC -> TRC -> CSV -> RLS.

RG status rule. Reality Grounding is Level 1 of the public method and a claim-authority precondition. It can force narrowing, escalation, exploratory-marking, or refusal, but it is not an option-rejecting gate; RF/NCRC, TRC, and CSV are the option-rejecting gates, and RLS ranks only what survives them.

Reality Grounding means the decision must say what part of reality it is actually touching before it claims what should be done. A decision sees only a surface, not the whole territory. Evidence grounds the surface. Unknowns mark the edge. Claims must stay inside the ground they stand on.

UCI/HOI placement in plain language. Containment is the structural safety step. UCI and HOI are structural sensors. If the sensors show containment-level damage, that belongs inside Containment. If the remaining options are tied or too close to call after Ripple Score, UCI/HOI may help choose the less brittle, less hollowing option, or the framework may refuse false certainty.

PLSS rule. PLSS is the zoom lens, not the safety waiver. It keeps local decisions practical while preserving rights, tail-risk, containment, and structural-diagnostic duties.

Reality is not whatever the majority believes, whatever an institution certifies, whatever a model outputs, whatever a dashboard computes, whatever a legal authority declares,

whatever is easiest to measure, whatever fits the preferred narrative, whatever produces the best score, or whatever the current surface happened to capture. Evidence is not reality. A score is not reality. A PCC is not reality. A model is not reality. Compliance is not correctness. Traceability is not truth. Plausibility is not evidence. Social force is not content-truth. Protection is not authority.

2.4 Why MathGov Does Not Claim Perfect Calculation

MathGov does not claim that all values, meanings, risks, harms, probabilities, or social effects can already be measured perfectly. Its purpose is more disciplined: to make assumptions visible, uncertainty explicit, rights floors non-negotiable, catastrophic risks harder to hide, and decisions easier to audit.

A number is not automatically truth. A score is useful only when the evidence behind it is visible, the uncertainty is disclosed, the affected stakeholders can be identified, and reviewers can challenge the assumptions. When the evidence is not strong enough, RippleLogic should not pretend certainty. It should narrow the claim, escalate for review, run sensitivity analysis, or refuse the stronger decision state.

This is why the system separates admissibility, permission, compliance, evidence, and execution. The goal is not perfect calculation; the goal is better judgment under declared constraints, with less hidden harm and more accountability to the full field of consequences.

Plain-language maturity boundary

The MathGov framework and RippleLogic are not a magic calculator. It does not make ethics automatic, and it does not remove human, legal, democratic, scientific, or professional responsibility. It makes ethical reasoning more structured, inspectable, rights-constrained, uncertainty-aware, and contestable.

Some parts are mature as architecture but still need testing as measurement tools. UCI, biological SGP measurement, calibrated kernels, reference calculators, validators, and empirical pilots remain future maturity work unless separately published and verified.

Section 3. Twelve Declared Primitives

A primitive is a foundational starting point that the system does not derive from anything earlier inside the system. A primitive is not a claim of absolute truth. It is a declared foundation, openly stated and open to challenge. The MathGov primitives are organized into three groups: foundational claims about reality (P1 to P4), foundational ethical commitments (P5 to P9), and foundational structural commitments (P10 to P12).

3.0 What MathGov Means by Reality

Root reality doctrine. MathGov defines reality, for decision-making, as the evaluator-independent field of constrained structures that, under transition, produce consequences. Structure is the stable arrangement of relevant differences. Constraint is what limits and

enables possible transition. Consequence is what follows when an action, event, policy, or decision enters that constrained structure. In public language: Reality is what exists, what constrains, and what follows.

Reality is named here operationally, not metaphysically. Consistent with the Canon's operational UBR boundary, MathGov does not assert a complete theory of being. It asserts something narrower and testable: decisions are embedded in an evaluator-independent structure of constraint and consequence that holds whether or not any agent perceives, admits, models, or has power over it. Evaluator-independent means independent of the evaluating agent's perception, admission, authority, model, or preference; it does not deny the reality of experience, social facts, institutions, norms, markets, laws, or model outputs when they shape welfare, rights, risk, agency, or available choices.

Root-science boundary. MathGov treats constrained structure under transition producing consequence as the operational substrate for governance-relevant reality. This does not claim that reality is only physical matter or that MathGov has final metaphysical access. It means that physical, biological, experiential, social, institutional, informational, artificial-system, and formal/logical structures and constraints become decision-relevant when they shape rights, risk, welfare, agency, available choices, legitimacy, or auditability.

A factor enters the reality surface of a decision when it bears on outcomes, welfare, rights, risk, containment, legitimacy, agency, or the structure of available choices. Constraint has at least two species: causal constraint, such as physical, biological, ecological, institutional, or social effects; and formal/logical constraint, such as coherence, conservation, computability, consistency, and mathematical possibility. MathGov uses both and must not collapse one into the other.

Three distinctions keep this primitive disciplined. First, vehicle, content, and effect must be separated: a belief, norm, institution, model, market signal, or legal status can be a real vehicle with real effects even when its content is false. Consensus can therefore be socially real and causally powerful while still failing to track reality. Agreement does not make a harmful system harmless. Second, surface and territory must be separated: every model captures only a reality surface, while the territory exceeds every surface. Omitted factors are recorded as unknown, not zero. Third, reality, truth, and value must be separated: reality is what holds, truth is a sufficiently faithful map, and value is what matters. P1 fixes the first and grounds, but does not replace, the ethical commitments that follow.

Experience is admitted to the reality surface in its own right, not solely because it may cause outward behavior. Pain, fear, attention, agency, dignity, and welfare-bearing experience are decision-relevant even when access to them is indirect and evidence-governed. SGP protection therefore does not depend on winning a contested claim that experience must be causally efficacious before it can matter.

Because the territory exceeds every modeled surface and audit operates only on the surface, no run can certify that all unmodeled catastrophic or containment-breaking effects are absent. Because catastrophic outcomes are, by the TRC definition, potentially irreversible or unbounded, folding them into an average computes over a tail finite knowledge cannot

reliably bound and misrepresents an uncompensable loss as compensable. This is why the non-averagable treatment of catastrophic risk and the CSV Gate follow from P1 under finitude rather than being optional severities added later.

Group A. Foundational claims about reality

P1. Reality Constrains Outcomes

Reality is not created by opinion, authority, ideology, wealth, status, institutional consensus, model output, or persuasive language. Those forces may be real vehicles with real effects, but their content can still be false. A policy can sound noble and still cause harm. A profitable system can still degrade the biosphere. A popular leader can still be wrong. A technically impressive AI system can still be misaligned. A social metric can rise while trust, health, ecological resilience, or institutional legitimacy collapses. Decisions have consequences whether or not anyone admits them.

Statement: Reality constrains outcomes, and decision systems must remain accountable to real consequences under declared uncertainty.

P2. Decisions Create Ripples

Every decision produces effects beyond the immediate actor. Effects may be direct or indirect, short-term or long-term, visible or hidden, local or systemic. This is the foundation of RippleLogic.

Statement: Actions propagate through interconnected systems, producing measurable and partially uncertain ripple effects.

P3. Stakeholders Exist Within Nested Unions

No entity exists in complete isolation for governance-relevant domains. Persons, institutions, technologies, animals, ecosystems, and future intelligences exist within nested patterns of interdependence. The canonical operational stack used in RippleLogic Tier 1 to Tier 3 computation is U1 Self, U2 Household, U3 Community, U4 Organization, U5 Polity, U6 Collective Managing Intelligence Union (CMIU, also called Humanity), U7 Biosphere. Two further layers, U8 Cosmic and U9 Universal / All-Encompassing Infinite Union (AIU), are recognized as non-computational meta-unions and serve only as humility horizons and precautionary amplifiers under the current release line.

Statement: All decision-relevant beings and systems exist within nested unions of relationship and consequence.

P4. Welfare Is Multidimensional

Well-being cannot be reduced to money, pleasure, productivity, GDP, or survival alone. The current canonical welfare dimensions are D1 Material, D2 Health, D3 Social, D4 Knowledge, D5 Agency, D6 Meaning, D7 Environment. These are operational categories, not metaphysical absolutes; they are open to refinement under governed update.

Statement: Welfare must be evaluated across multiple dimensions, not collapsed into a single crude metric.

Group B. Foundational ethical commitments

P5. Rights Floors Are Non-Compensatory

Some harms cannot be traded away because aggregate benefit appears positive. A system must not be able to say: most people benefit, so this group can be exploited. This is the basis of the Non-Compensatory Rights Constraint (NCRC). Welfare scoring may rank admissible options; it may not override rights floors.

Statement: Certain rights and protections function as floors, not negotiable preferences.

P6. Catastrophic Risk Cannot Be Averaged

Some risks are too severe, too irreversible, or too civilization-threatening to be treated as ordinary costs inside an expected-value calculation. A small probability of catastrophic failure still matters when the downside is structurally irreversible. This is the basis of the Tail-Risk Constraint (TRC), implemented through Conditional Value-at-Risk (CVaR) over a governed scenario set.

Statement: Catastrophic and irreversible risk must be bounded, not averaged.

P7. Larger Systems Constrain Smaller Ones

A system can win locally while damaging the larger system it depends on. A company can profit while destroying worker health. A nation can grow while degrading the biosphere. An AI lab can advance capability while raising alignment risk. This is the basis of the CSV gate.

Statement: No system may optimize one layer of reality by destabilizing the larger systems that sustain it.

P8. Moral Consideration Scales With Sentience

A rock, a bacterium, a tree, a forest, a dolphin, an elephant, an orangutan, a human, and a future artificial intelligence do not occupy identical ethical positions. Moral consideration should not be based on species membership, power, familiarity, or economic convenience alone. The Sentience Gradient Protocol assigns protection-relevant moral status under uncertainty according to evidence of awareness, agency, welfare/valence capacity, and union-participation capacity. Within the current canonical line, the human plateau (SG_norm(H) = 1.0) is treated as absolute; class-level and entity-level plateau assignments outside humans follow governed SGP procedure.

Statement: Moral consideration is assigned according to evidence of sentience and welfare-bearing capacity, under uncertainty, with the human plateau treated as absolute.

P9. Domination Is a Distinct Harm

The Minimal Normative Axiom of Canon Section 3.1 centers not only suffering but domination. Concentrated, unaccountable, or coercive power over others can degrade dignity, agency, and the conditions for flourishing even where measurable welfare appears unchanged. Anti-capture, non-domination, and the preservation of meaningful contestability are first-order concerns of the framework.

Statement: Reducing unnecessary suffering and reducing domination are co-equal normative concerns; neither can be traded for the other.

Group C. Foundational structural commitments

P10. Moral Status Is Not Decision Authority

Four separations are constitutive of the framework and must not be collapsed: (i) moral status is not decision authority; (ii) protection is not governance role; (iii) sentience is not the same as intelligence; (iv) admissibility logic is not welfare optimization. These separations matter because each pairing is regularly conflated in real institutional practice, and that conflation is one of the routes by which rights, audit, and containment are eroded.

Statement: Moral status, decision authority, protection, governance role, sentience, intelligence, admissibility, and optimization are distinct objects and must not be collapsed.

P11. Uncertainty Must Be Made Visible

Most real decisions involve incomplete evidence, model uncertainty, value disagreement, measurement error, unknown side effects, and strategic behavior. The danger is not uncertainty itself; the danger is hidden uncertainty pretending to be certainty. RippleLogic requires uncertainty to be represented openly through declared methods, sensitivity analysis, and audit-flag semantics.

Statement: Uncertainty must be represented openly rather than buried inside authority, rhetoric, or false precision.

P12. Power Must Be Auditable

Human authority alone is not stable enough at civilization scale because humans and institutions can be corrupted, coerced, captured, bribed, threatened, gamed, or distorted by incentives. The remedy is not to remove humans from governance; the remedy is to constrain authority by transparency, evidence, declared procedure, and contestability. The Provenance and Compliance Certificate (PCC), the Five-Sentence Public Rationale (5SPR), and the Notice-Choose-Act-Reflect (NCAR) loop exist to make legitimate authority inspectable.

Statement: Legitimate decision power must be inspectable, challengeable, evidence-linked, rights-constrained, and accountable over time.

Section 3A. Primitive Dependency Audit

The twelve MathGov primitives are declared starting points for this framework. They are not claims of absolute metaphysical irreducibility. Each primitive depends on background concepts, evidence traditions, measurement practices, and normative choices that must remain visible.

This matters because a decision framework becomes more trustworthy when it declares where it begins, what it imports, what it assumes, what it measures, and what could force revision.

The following audit posture applies to all twelve primitives.

1. A primitive is a declared starting point inside MathGov, not an assertion that no deeper analysis is possible.
2. A primitive may be imported from parent disciplines such as systems science, ethics, law, risk science, cognitive science, ecology, political theory, or decision theory.
3. A primitive remains legitimate only if its dependencies are disclosed and its operational use remains testable, contestable, and revisable under governed update.
4. A primitive should not be protected by reinterpretation after reality contradicts its projected consequences.
5. Where a primitive cannot generate a unique operational consequence for a run, the correct response is not false certainty. The correct response is refusal, escalation, or bounded provisional action under declared uncertainty.

Primitive dependency checklist:

For each primitive P1 through P12, reviewers should ask:

- What does this primitive presuppose?
- What parent disciplines does it import from?
- What operational role does it play in RippleLogic?
- What would count as evidence that this primitive is incomplete, misframed, or misapplied?
- Does this primitive affect admissibility, scoring, interpretation, or public communication?
- Does the current run depend on this primitive in a way that should be disclosed in the PCC?

Plain-language rule. MathGov does not become stronger by pretending its primitives float above all prior thought. It becomes stronger by declaring its starting points, exposing their dependencies, and letting reality test their consequences.

Summary statement. The MathGov primitives are declared foundations for an ethical decision architecture. They are not final metaphysical proof. They are disciplined commitments that must remain visible, testable, and open to governed refinement.

Section 3B. The Reference Stack: Against What Is the Decision Being Judged?

A decision system is not trustworthy simply because it has rules, dashboards, audits, or guardrails. Those tools matter only when they are connected to a valid reference.

RippleLogic asks a simple but demanding question: Against what is this decision being evaluated?

For a bridge, the reference may include load, stress, gravity, material strength, and failure thresholds. For governance, the reference is more layered. It includes bodies, needs, rights, ecosystems, institutional legitimacy, causal evidence, uncertainty, and audit records.

RippleLogic does not pretend that every governance question has one simple physical reference. Instead, it uses a reference stack: physical constraints; biological needs; rights floors; catastrophic-risk corridors; containment conditions; stakeholder impacts; governance authority; evidence status; and auditability.

This is why The MathGov framework and RippleLogic are not merely a scoring system. It is a reference-structured decision architecture. If the reference is missing, circular, unfalsifiable, or too weak for the claim being made, the honest answer is not forced selection. The honest answer is refusal, escalation, more evidence, or a narrower claim.

Plain-language rule. Do not govern by assumption. Declare the reference, show the evidence, disclose the uncertainty, and refuse false certainty.

Section 3C. Category Grounding: Are the Words Still Attached to Reality?

Reality Grounding asks what part of reality the decision has actually touched. Category Grounding asks whether the categories used to describe that reality are bounded correctly enough for the claim being made.

This matters because a wrong category can make a good-looking metric dangerous. If the category is wrong, the metric may measure the wrong thing. If the metric measures the wrong thing, optimization can improve the wrong target. If governance protects the optimized metric, the system may become increasingly efficient at maintaining drift.

Simple example. If a school optimizes “learning” but defines learning only as test-score improvement, it may raise the metric while weakening curiosity, confidence, teacher trust,

long-term understanding, and student agency. Category Grounding asks whether “learning” was defined correctly before the score is optimized.

Category Grounding is not a sixth step. It belongs inside Reality Grounding when material. The five-level method remains: Reality Grounding -> Rights Floor -> TRC -> CSV -> RLS.

A category is material when it affects who counts as a stakeholder, what counts as harm, what counts as sentience or agency, what counts as safety or stability, what counts as catastrophic risk, what counts as containment damage, what counts as authority, or what metric will be optimized.

Reviewers should ask: What does this term mean? What boundary defines it? What belongs inside it? What does not belong inside it? What discriminator separates it from nearby terms? Where does it sit in the dependency order? What reference structure tests it? What evidence surface supports it? What would force revision or refusal? What happens if this category is wrong?

Plain-language rule. Before MathGov scores a thing, it checks whether the thing has been named, bounded, and grounded well enough for the decision. A score is not a cure for a bad category.

Semantic debt. Semantic debt is the cost of continuing to use categories whose reality correspondence has become weak, stale, inherited, contested, or unreviewed. MathGov does not become stronger by hiding semantic debt. It becomes stronger by naming it, narrowing claims, and revising categories when reality requires it.

Section 3D. Structural Viability: Can the Plan Actually Stand?

After a decision is grounded, rights-safe, tail-risk bounded, and containment-safe, one more question may matter before scoring or execution: can this plan actually stand? Structural Viability Check asks whether the option has the resources, dependencies, reversibility or endpoint clarity, operational capacity, and internal consistency required for current execution.

This is not a sixth public step. It belongs inside Containment when execution feasibility is material. The public method stays: Ground -> Protect -> Bound -> Contain -> Score.

Plain rule: a good intention is not enough if the plan cannot structurally work. But structural viability is not justice either. A plan can be stable and still violate rights. Rights floors remain first and non-negotiable.

A structurally non-viable option may still be kept as a redesign target. It should not be presented as currently selectable, executable, deployment-ready, or deterministically selected.

Section 3E. No Silent Remainder: Unknown Is Not Zero

MathGov allows uncertainty. It does not allow silent blanks. A required field may be estimated, assumed, declared unknown, not applicable, or blank by design. It may not simply disappear. Silent gaps are how false certainty enters a decision.

Plain rule: if the decision depends on it, name it. If you do not know it, say you do not know it. If it does not apply, say why. Do not turn unknown into zero.

Section 4. Union as a Reality Principle

Union does not mean everything is the same. In MathGov, it means that decision-relevant beings and systems exist through relationships, dependencies, and consequences.

This is the core of what Union-Based Reality (UBR in the canonical specification) describes. A human being is distinct, but never isolated. The body is built from organs, cells, water, minerals, breath, food, ancestry, language, culture, and environment. A family is built from individuals. A society is built from families, institutions, laws, ecosystems, and shared meanings. A biosphere is built from interdependent life. Every intelligence or intelligence-like system we have actually observed, biological, social, or artificial, operates through relationships among memory, perception, feedback, environment, and context. We have never found a real system in nature that operates in true isolation, and current scientific understanding gives us little reason to expect one.

That is what union means in this framework. Not a doctrine, not a slogan, not a feeling. It is the operational condition that decisions happen inside structured relationships, not inside isolated abstractions.

Union does not erase difference. Water is not fire. A human is not a dolphin. A tree is not a government. A child is not an artificial intelligence. Distinctions matter. Boundaries matter. Rights matter. The framework's commitment to non-compensatory rights floors only makes sense because individual beings are real and their integrity is real.

But distinction is not absolute separation. Every being and system exists within larger fields of dependency and consequence. Actions ripple. Care propagates. Harm propagates. Trust propagates. Collapse propagates.

These do not propagate at the same rate. Reality is not symmetric. A drop of poison ruins clean water; a drop of clean water does not redeem poisoned water. Construction takes years and destruction takes minutes. Trust is built slowly and lost quickly. Health degrades faster than it heals. A lie travels around the world before the truth has its boots on. Union is not balance in the sense of equal harm and equal benefit. Reality is asymmetric, and any honest system has to say so.

This rules out three common misreadings of union, all of which MathGov rejects.

The first is cosmic merger: everything is one, so distinctions and harms do not really matter. This dissolves rights, accountability, and the moral weight of suffering, and it usually arrives wrapped in beautiful language. The framework rejects it.

The second is required opposition: every good must be matched by an equal bad to keep things in balance. This is a misuse of yin and yang, of karma, of dialectic, and of the rhetoric of “both sides.” A person does not need to do harm to do good. A system does not need to produce equal damage and benefit to be healthy. Clean water sustains life better than poisoned water. Truthful information serves people better than deception. Health is better than preventable sickness. Cooperation is more stable than domination over the long run. Better balance does not mean equal parts harm and benefit. Better balance means right relationship.

The third is collectivist dissolution: the individual is only a fragment of the whole, so individual rights bend to collective will. This is the totalitarian misuse of unity, and it is precisely the failure mode the rights gate (NCRC) is built to refuse.

Union also does not flatten attention. A breakfast choice acts strongly on the self, weakly on the household, and almost not at all on the polity. A national infrastructure decision acts the other way around. The discipline is to match moral and analytical attention to actual coupling, dependency, and consequence, not to perform equal concern at every scale.

So the MathGov claim is not “everything is one, therefore nothing matters.” The claim is the opposite. Things have real identities, and those identities exist inside networks of cause, dependence, and effect. Decisions have to be evaluated by their ripples across the systems they actually touch.

Different vocabularies converge on the same observation. Things are connected. Connection has structure. Structure has consequences. Consequences are not symmetric. Decisions matter.

Union is not dogma. In the operational sense used here, union is the condition that every decision unfolds inside a field of relationship, dependency, consequence, and accountability.

Section 5. Eleven Declared Assumptions

An assumption is a provisional claim used to make the system runnable. Assumptions are weaker than primitives. They must remain visible, challengeable, and revisable as evidence and practice accumulate.

A1. The seven welfare dimensions are a strong starting set

Material, Health, Social, Knowledge, Agency, Meaning, and Environment appear broad enough to capture most major welfare effects relevant to current governance. They may be refined, subdivided, or expanded under governed update. They are not claimed to be metaphysically final.

A2. The seven operational unions are sufficient for most current decisions

U1 Self through U7 Biosphere are sufficient for most contemporary governance, policy, business, AI, and ethical decisions. The Cosmic and Universal layers should remain humility horizons, not over-quantified before evidence supports parameterization.

A3. Rights floors can be specified well enough to guide decisions

Societies can identify core protected domains: life, bodily integrity, basic needs, dignity, liberty, due process, information integrity, and ecological integrity. Disagreement at the edges does not erase the reality of obvious violations; the existence of hard cases does not eliminate clear ones.

A4. Sentience can be evaluated on a gradient under uncertainty

Moral status is not all-or-nothing. Evidence of awareness, welfare/valence capacity, agency, self-modeling, learning, communication or species-appropriate behavioral markers, and union-participation capacity can support graduated estimates. SGP procedures are designed to make these estimates governed, auditable, and corrigible rather than ad hoc. The human plateau is treated as absolute.

A5. Weights can be made legitimate through Hybrid Democratic Weighting

Welfare weights become more legitimate when they combine constitutional floors, stakeholder input, expert evidence, democratic review, sensitivity testing, and public audit. Hybrid Democratic Weighting (HDW) is the canonical mechanism for this combination. PLSS, where declared, governs prominence-adjusted residual welfare ranking under the same anti-capture logic.

A6. Quantification can improve ethics if it remains humble

Not everything valuable can be perfectly measured. Structured estimation is often better than unstructured power. The danger is false precision; scores must be treated as decision aids, not divine commands. RippleLogic addresses this through layered cascade, audit flags, and the prohibition of Tier-4 claims absent ProofPack.

A7. Open systems are less capture-prone than closed systems

Open standards, public documentation, audit trails, and reproducible decision records reduce capture risk. Open does not guarantee integrity, but closed systems are easier to monopolize, manipulate, and hide.

A8. Better decision architecture can reduce capture and authoritarianism

MathGov does not claim to defeat authoritarianism, corruption, or destabilizing power concentration. It aims to make these failure modes substantially harder to sustain by forcing decisions into visible structures: who is affected, what rights are at stake, what harms are

hidden, who benefits, who bears the risk, what evidence supports the decision, and whether the reasoning can be reconstructed later.

A9. AI and digital systems will increasingly participate in governance

Whether or not current AI is sentient, AI systems already influence decisions: ranking information, automating workflows, recommending actions, detecting patterns, shaping perception, and mediating institutional behavior. Governance architecture must account for AI as both a tool used by decision-makers and a possible future stakeholder if SGP-grade evidence of moral patienthood emerges.

A10. The system itself can be gamed and must model agents

Any governance method can be captured, bureaucratized, moralized, weaponized, ignored, ritualized, or turned into ceremonial rationality. The framework must therefore model agents and incentives: who benefits, who resists, who can manipulate the process, who can route around it, and who bears the risk when the system fails.

A11. Corrigibility must hold, with a floor on what is revisable

MathGov must remain corrigible, evidence-responsive, and open to structural improvement, otherwise it ossifies into dogma. At the same time, corrigibility itself requires a floor: the rights gate, the catastrophic-risk gate, the CSV gate, and the auditability commitment cannot be revised away by ordinary stakeholder process. Revision of these floors is constitutional in character and requires governed extension protocol, not ordinary update.

Section 6. What MathGov Does Not Assume

Equally important to what MathGov declares is what it refuses to claim. The framework explicitly does not assume:

- That it possesses final ontology or final moral truth.
- That all values reduce perfectly to numbers.
- That humans should surrender governance to AI systems.
- That current AI systems are conscious or moral patients.
- That all beings have equal moral status.
- That powerful beings matter more than vulnerable beings.
- That a majority can vote away fundamental rights.
- That data is neutral or self-interpreting.
- That models cannot be captured.
- That formalization eliminates moral conflict.

- That one culture, nation, species, or institution should define flourishing for all others.
- That a Tier-4 conformance claim is currently legitimate (it is prohibited until ProofPack is publicly replayable and independently verified).

This boundary statement exists because the framework must avoid becoming exactly what it is designed to resist: a rigid authority system pretending to be universal while concealing its own assumptions.

Section 7. The Operational Cascade

RippleLogic operates as a five-level admissibility-and-ranking method. Each level answers one distinct question. Levels do not collapse into a single blended score; later levels do not override earlier levels; earlier success does not excuse later failure.

Level 1, Reality Grounding / Reality-Surface Grounding. What reality surface, evidence trace, material unknowns, category boundaries, transition boundary, consequence pathways, and claim boundary support the run? If the claim exceeds its reality surface, the run must narrow, escalate, gather evidence, mark the result exploratory, or refuse the stronger claim.

Level 2, Rights Floor / NCRC. Does this option cross a non-compensatory rights minimum? An option that does is inadmissible. Welfare ranking cannot rescue a rights-floor violation.

Level 3, Tail-Risk Bound / TRC. Among rights-admissible options, is the catastrophic downside acceptably bounded? An option whose CVaR over the governed scenario set exceeds the declared corridor is inadmissible, regardless of expected welfare.

Level 4, Containment. Among admissible options, does this option preserve sufficient integrity of the larger system that contains the local gain? An option that fails Containment is not selectable, even if it ranks highly on residual welfare.

Level 5, Ripple Score / RLS. Among selectable options, which produces the best residual welfare pattern under the declared weighting regime? RLS ranks; it does not rescue failures in Reality Grounding, NCRC, TRC, or CSV.

Special-case UCI / HOI tie-break. UCI and HOI are structural diagnostics used inside Containment when material. Only after options are already in the selectable set may residual UCI/HOI help document a tied, close, uncertainty-overlapped, or non-decisive RLS result. They do not reopen earlier gates and they are not a substitute welfare score.

The non-overlap rule is canonical: Reality Grounding = claim authority; NCRC = rights feasibility; TRC = catastrophe feasibility; Containment = cross-scale structural feasibility; RLS = residual welfare ranking within the selectable set; residual UCI/HOI = non-decisive structural resolution or monitoring within the selectable set. No implementation, commentary, or downstream artifact may collapse these functions into a single blended score.

Section 8. The Five Decision States

RippleLogic recognizes five distinct decision states. They must not be collapsed, and a later state must not be used to infer an earlier one.

Rights-admissible. The option passes NCRC.

Admissible. The option passes NCRC and TRC.

Selectable. The option is admissible and passes Containment (Mode A).

Selected. The option is chosen from the selectable set after RLS, tie-break, and any required escalation disposition.

Refusal. The framework cannot uniquely determine the applicable admissibility or selection state from the declared inputs, evidence, projection rules, or layer commitments.

No downstream score may rescue failure at an earlier state. A welfare gain cannot make a rights-violating option admissible. A high RLS cannot make a containment-failing option selectable. Refusal must not be silently converted into gate failure, gate success, or discretionary selection. The decision-state vocabulary exists precisely to prevent these substitutions.

8.1 What refusal means in plain language

Refusal means the framework cannot honestly make the claim being asked of it. It is not the same as rejecting an option on ethical grounds. It is not the same as selecting the safest-looking option by instinct. It is a statement that the available evidence, declared projection, or category structure is insufficient for the requested verdict.

For example, if a decision lacks enough evidence to determine whether a protected subgroup is harmed, the correct result is not to assume safety. If a scenario model cannot determine whether catastrophic risk is inside the corridor, the correct result is not to average the uncertainty away. If a sentience claim relies only on fluent language or self-report, the correct result is not to treat that as proof of moral patienthood.

Refusal can lead to more evidence gathering, redesign, escalation, a narrower claim, or bounded provisional action where delay itself would create harm. What it may not do is pretend that uncertainty has been solved.

Section 9. Critical Separations

Four separations are constitutive of the framework and merit explicit restatement, because each is regularly conflated in institutional practice and each conflation opens a route to rights, audit, or containment failure.

Moral status is not decision authority. A being can warrant moral consideration without thereby acquiring governance role. Authority is grounded in responsibility, accountability, and procedural legitimacy, not in moral status alone.

Protection is not governance role. A protected stakeholder is not, by virtue of protection, a decision-maker over others. SGP confers grounds for protection; it does not confer authority.

Sentience is not intelligence. Capability and welfare-bearing capacity are distinct. A highly capable system may not be a moral patient; a moral patient may not be highly capable. SGP scoring tracks the second; capability evaluation tracks the first.

Admissibility is not optimization. RF/NCRC, TRC, and CSV are feasibility gates. RLS is welfare ranking. They have different inputs, different operators, and different failure consequences. A welfare maximum that is not admissible is not selectable; an admissible option with mediocre RLS is still selectable.

Section 10. SGP Integration: Plateau, Scalar Binding, Authority Boundary

Current Human Plateau / CMIU clarification. Humans are not treated as full-plateau beings because of arbitrary species preference. Human persons receive non-degradable protection so rights cannot become a capacity test, and humanity currently provides the evidenced calibration anchor for reality-managing intelligence within the Collective Managing Intelligence Union (CMIU). This does not make humans metaphysically superior, and it does not close the plateau to non-human biological or artificial entities that meet the governed standard.

SGP v6.4 is the canonical companion specification for moral status, rights of protection, graduated moral consideration across substrates, and moral-status term discrimination. Three SGP commitments matter most for foundational understanding.

10.1 The human plateau is absolute

Human moral status is treated as plateau-bearing with $SG_norm(H) = 1.0$. This plateau is not subject to downward revision through ordinary update procedure. Class-level plateau assignments for non-human entities follow governed SGP class-level plateau records; entity-level plateau assignments follow governed individual plateau records. Plateau status is not inferred informally from taxon evidence alone.

10.2 SGP scalar binding: welfare stream only

SGP-derived welfare-stream scaling (s_k) is applied only in the welfare stream used for RLS ranking, using $SG_patient_norm(E)$ where available and otherwise $SG_norm(E)$, while preserving the Human Plateau Rule and leaving admissibility layers unchanged. Admissibility layers (NCRC, TRC, CSV) operate on the base stream with s_k set to 1. This separation is non-negotiable: SGP weighting cannot be used to lower a rights floor, soften a

catastrophe gate, or weaken a containment check. SGP shapes residual welfare comparison among already-feasible options; it does not shape feasibility.

10.3 The authority boundary

SGP confers grounds for protection and for inclusion in welfare and rights computation. SGP does not confer decision authority over other stakeholders, governance role, voting weight, or institutional power. The separation between moral status and authority is constitutive: a being can be highly protected and have no governance role, and an entity can hold legitimate authority while being morally considerable on the same terms as everyone else.

Section 11. The Operating Sequence

A RippleLogic decision proceeds through eleven steps. Tier requirements (Section 15 of Canon) determine which steps are mandatory at which level of rigor.

1. Define the decision. State what is being decided, who is deciding, what options exist, and what happens under no action. Declare tier and configuration.
2. Map stakeholders across affected unions. Identify the relevant scope chain from U1 Self through U7 Biosphere; note where U8 Cosmic or U9 Universal / All-Encompassing Infinite Union (AIU) humility considerations apply.
3. Identify sentient and welfare-bearing stakeholders under SGP. Apply governed SGP procedure for class-level and entity-level moral status and rights of protection.
4. Apply the Rights Floor (formal NCRC). For each option, check whether any rights-covered cell crosses a non-compensatory minimum on the worst-off protected subgroup. Eliminate any option that fails.
5. Apply TRC. For each rights-admissible option, evaluate CVaR over the governed scenario set against the declared corridor. Eliminate any option that exceeds the corridor; record TRC fallback if invoked.
6. Apply Containment (Mode A where required). For each admissible option, evaluate cross-scale structural feasibility; eliminate or escalate any option that degrades a containing union beyond tolerance.
7. Score residual welfare via RLS. Rank selectable options across the active 49-cell welfare matrix using HDW (or PLSS, where declared) weights and declared uncertainty method.
8. Make weights, evidence, and uncertainty visible. Declare union weights, dimension weights, evidence quality, stakeholder input, expert adjustments, uncertainty bands, and sensitivity tests in the PCC.
9. Model agents and incentives. Identify who might resist, game, distort, or route around the decision; record relevant audit flags and mitigation plans.

10. Produce an auditable decision record. Emit the Provenance and Compliance Certificate (PCC) with the Five-Sentence Public Rationale (5SPR) and any required containment, SGP, and stewardship disclosures.
11. Run NCAR. Notice, Choose, Act, Reflect. After implementation, evaluate outcomes, surface missed harms, revisit weights, and trigger governed update where evidence requires it.

Section 11A. A Simple Reading of the System

A simple act can still be read through the full architecture without overcomplication. Suppose a parent has money to buy a toy for their own child and notices another child nearby who has less. The parent could buy for their own child, buy for the other child, buy something shareable, save the money, or do nothing.

MathGov does not command one sentimental answer. It asks for disciplined attention. Are any rights at stake? Is anyone being coerced, neglected, humiliated, or endangered? Is there any catastrophic risk? Usually not. Is there a containment issue, such as undermining the parent's household obligations or creating unhealthy dependency? Usually minor, but still worth noticing. Then the decision moves into residual welfare: self, household, nearby community, social trust, meaning, and material cost.

In a low-stakes moment, the full calculation can be mental and lightweight. In a high-stakes policy decision, the same logic becomes formal, documented, and auditable. This is the point of tiers: the structure is stable, but rigor scales with consequence.

The deeper lesson is that RippleLogic is not only for governments or AI systems. It trains attention. It asks: who is touched, what is protected, what could go wrong, what improves, what is uncertain, and what should be learned after action?

Section 11B. Execution Stability in Plain Language

A decision is not finished merely because an option can be described, scored, or technically executed. Before a consequence-bearing action becomes real, RippleLogic now asks one additional practical question: what kind of transition are we about to enter?

The Execution Stability Diagnostic classifies the transition itself. It asks whether the move preserves viable continuation, requires monitoring, creates bounded destabilization, intentionally closes an endpoint, fails containment, or remains underdetermined. This helps prevent a system from treating execution as automatic just because a draft, instruction, or score exists.

This does not make stability the highest moral value. Some stable systems are unjust. Some necessary reforms temporarily disturb existing arrangements. RippleLogic therefore keeps the order clear: rights floors, catastrophic-risk bounds, containment, and legitimate authority

come first. Stability is a diagnostic about the transition into consequence, not a permission slip to preserve oppression or avoid needed repair.

Plain-language rule. Do not ask only, can this be done? Ask: should this transition become real, under this evidence, with these affected stakeholders, this reversibility profile, this monitoring plan, and this claim boundary?

Simple example. If an institution wants to deploy a new AI assistant, the ordinary cascade asks whether rights are protected, tail risk is bounded, containment is preserved, and welfare improves. The execution stability check then asks whether deployment is reversible, whether failure modes are monitored, whether users have recourse, whether the system can be rolled back, and whether any endpoint-like consequence is being created without proper authority.

Section 12. Posture: Structured Humility Under Reality

A decision is better when it protects rights, bounds catastrophic risk, preserves the integrity of containing systems, improves welfare across affected unions, makes uncertainty visible, preserves auditability, and remains open to correction.

More compactly: choose the action that creates the best defensible ripple across affected stakeholders without violating rights and without destabilizing the systems that make life possible.

The deeper posture of MathGov is not final truth by declaration. It is structured humility under reality. The framework declares its primitives, exposes its assumptions, protects its rights floors, models uncertainty, documents reasoning, and remains open to correction as better evidence and better models emerge.

The goal is not perfect morality by calculation. The goal is to make ethical and governance decisions harder to corrupt, harder to hide, harder to capture, and easier to test against reality, so that intelligences (human, institutional, and eventually artificial) can align with reality, with rights, with long-term survival, and with the flourishing of interconnected life itself.

Section 13. What v11.4 Integrates and Completes: Scoring Literacy, Stream Separation, SGP v6.4, Structural Viability, Term Discrimination, and Release Boundaries

Current completion note. Core Release 2026.09 contains RippleLogic Canon v11.4, SGP v6.4, ripple.md v4.4, Agent System v11.4, Primer v3.4, and Aligners Sheet v4.4. The line preserves the core cascade while adding decision-verdict hardening, openability repair, and the Reality Grounding and Lean Cascade Patch: structure-transition-consequence vocabulary, vehicle/content/effect separation, surface/territory discipline, status-vs-trace audit flags, and non-averagability derivation under finitude.

The v10.1–v10.8 historical methodology hardening does not change the soul of MathGov or the core RippleLogic cascade. It makes the scientific posture stricter. A high-stakes claim should not merely sound coherent, fit the past, or be compatible with observed outcomes. It should declare what would count against it before the evidence is used to protect it.

The plain-language rule is simple: say what you are assuming, say what follows if the assumption is right, say what we should observe, say how much error is tolerated, and say when the claim must be refused, downgraded, or abandoned.

13.1 The seven-step trace

Primitive commitment → Operator → Regime → Projection → Observable consequence → Tolerance → Verdict.

13.2 What this prevents

This prevents post-hoc rescue, category collapse, symbolic overconfidence, vague claims of validation, and the habit of calling a useful model proven. It also prevents the opposite error: refusing to act where a bounded, rights-preserving, reversible, and well-audited provisional action is justified.

13.3 What this does not claim

The carried-forward v10.8 method posture does not claim final truth, perfect objectivity, complete prediction, or immunity from formal limits. It says something more practical: every serious claim must carry its own audit trail, evidence boundary, and reality-facing test.

13.4 Binary Gates and Uncertain Evidence

MathGov and RippleLogic distinguish two ideas that are often confused. A decision gate can be binary, while the evidence used to evaluate that gate can still be uncertain.

For example, a rights floor may pass or fail. A tail-risk corridor may be exceeded or not exceeded. A containment relation may hold or fail. But the evidence used to evaluate these option-rejecting gates can involve measurement error, incomplete models, probability estimates, contested data, or human judgment.

The carried-forward v10.8 rule is simple: do not pretend uncertainty is gone merely because the decision surface needs a yes/no answer. Binary gates make governance accountable. Evidence humility keeps the binary answer honest.

When evidence is not strong enough to support the requested claim, RippleLogic should not force a confident decision. It should narrow the claim, request more evidence, escalate to review, or refuse the stronger claim. Refusal is not failure; it is the system refusing to turn uncertainty into false certainty.

This is why high-stakes decisions must identify the transition being proposed, the affected scopes, the primitive assumptions, the projection path, the observable consequences, and the conditions that would trigger rollback or refusal before consequence-bearing execution.

Section 14. What Is Mature Now, and What Still Requires Validation?

Current maturity note. Canon v11.4 is complete as the current core specification for this release line. SGP v6.4 is complete as the current moral-status and protection protocol. ripple.md v4.4, Agent System v11.4, Primer v3.4, and Aligners Sheet v4.4 are synchronized companions. Biological SGP measurement remains guidance-stage rather than measurement-complete, and stronger empirical or ProofPack claims remain non-claimable without separate evidence.

RippleLogic is mature as an architecture and decision methodology for its declared Tier 1-3 scope, but not every measurement surface is equally mature. This distinction protects the framework from overclaiming and helps reviewers understand where deployment caution is required.

14.1 Mature now

The cascade architecture is specified: rights floors first, then tail risk, containment, residual scoring, and structural diagnostics.

The claim-boundary language is explicit: GitHub-ready does not mean ProofPack-ready, Tier 4-ready, or empirically validated.

The audit vocabulary is strong enough for worked-run disclosure and reviewer challenge.

14.2 Still requiring validation

UCI is load-bearing, but cross-domain validated UCI instruments are not yet published.

Kernel propagation is design-ready, but domain-calibrated and backtested kernels remain future work.

SGP biological evaluation is protection-useful, but individualized biological measurement requires species-appropriate annexes and expert validation.

HDW can expose weights, but institutional capture must be actively audited rather than assumed away.

PLSS keeps local decisions practical, but escalation triggers must be visible so local convenience does not hide non-local harm.

14.3 Plain-language rule

The framework may be structurally ready before every instrument is fully validated. Good governance requires both: use the architecture now where appropriate, but disclose the maturity of the evidence, measurements, kernels, institutions, and escalation mechanisms behind each claim.

14.4 Cross-Artifact Concept Map

RippleLogic Canon v11.4 is the governing artifact. It defines the cascade, equations, admissibility gates, PCC requirements, claim boundaries, conformance vocabulary, reality-grounding records, schema ownership boundaries, and machine-readable registry expectations.

SGP v6.4 is the companion moral-status and protection protocol. It provides welfare-relevant protection scalars and evidence objects. It does not grant governance authority and does not override RippleLogic admissibility gates.

ripple.md v4.4 is the portable alignment contract and assurance wrapper. It helps implementations record evidence, decision notes, update legitimacy, reality-grounding fields, refusal conditions, and deployment assurance. It does not replace the Canon.

RippleLogic Agent System v11.4 governs agent deployment, security posture, mode control, operator authority, audit logging, reality-reference closure, refusal, rollback, and runtime conformance. It does not authorize an agent to bypass the Canon.

Ripple Aligners Sheet v4.4 is a worked-run exemplar and training companion. It helps users inspect and replay a sample run, including FrameworkVerdict and RealityGroundingStatus surfaces. It is not a ProofPack, semantic validator, or reference calculator.

Foundations Primer v3.4 is explanatory and carries no normative force. It helps readers understand MathGov and RippleLogic in plain language.

Release-integrity files document the release package. They verify file integrity and packaging claims, but they do not provide empirical validation or ProofPack readiness.

Machine-readable registries, schemas, CI checks, reference calculators, and calibration packs are tooling-line ProofPack-prep surfaces. Unless separately listed in the release manifest, they are not bundled in the v11.4 core foundation release. They improve machine-checkability and replay readiness but do not by themselves create Tier 4 readiness.

Foundations Primer v3.4 is informative and carries no normative force. This release is not an empirical-validation claim, not ProofPack-ready, not machine-verifiable-ecosystem-complete, and not Tier 4-ready. Stronger claims require separate schemas, validators, reference calculators, replayable PCC artifacts, calibration evidence, and independent replay instructions.

14.5 What Changes in Practice

A RippleLogic-informed decision process changes institutional practice in five concrete ways.

First, rights floors are checked before welfare optimization. A high-scoring option cannot rescue a rights violation.

Second, catastrophic tail risk is checked before ordinary ranking. Expected benefits cannot erase unacceptable catastrophic exposure.

Third, local gains must be tested against containing systems. A decision that helps a local actor while degrading a household, institution, polity, humanity, or biosphere-level support system requires escalation or refusal.

Fourth, uncertainty is not hidden. Weak evidence, missing indicators, unresolved maturity limits, and underdetermined projections must be disclosed.

Fifth, the decision record becomes auditable. The system records what was considered, what was excluded, which evidence was used, which assumptions were made, which gates passed or failed, and why selection or refusal occurred.

Section 15. Practical Validation Path and Civilizational-OS Boundary

15.1 Civilizational OS as aspiration, not proven status

MathGov may be described as a proposed ethical operating system or civilizational operating-system candidate when speaking about its design ambition. That language should remain subordinate to the evidence. The current claimable status is more modest: the MathGov framework is a governance-and-alignment research program with RippleLogic as its ethical decision operating system with a partially calculable ethical decision architecture, an assurance wrapper, and an agent governance companion.

15.2 Protection is not participation

SGP is clearest when it separates moral patienthood from governance participation. A being can deserve protection even if it cannot deliberate, communicate, vote, govern, or carry stewardship responsibility. This is especially important for animals, infants, cognitively atypical humans, non-linguistic minds, and future artificial or hybrid systems.

15.3 Evidence labels for ordinary users

When MathGov displays a number or score, users should be able to see whether the value was measured, estimated, assumed, expert-judged, contested, unvalidated, not observed, or only a sensitivity placeholder. This reduces false precision and helps people understand what the system knows, what it guesses, and what it has not yet proven.

15.4 Compact profiles

For practical adoption, MathGov should be offered through compact profiles: policy review, procurement review, AI deployment review, university AI policy review, and family/community decision support. These profiles can simplify the process, but they cannot waive rights floors, catastrophic-risk checks, containment, refusal, or public accountability.

Section 16. Historical Lineage: Why v10.8 Focused on Integration, Not Grand Claims

The next meaningful improvement is not a larger claim about civilization. It is better integration: making sure evidence labels, SGP protection boundaries, compact profiles, workbook sanity checks, and release verification all point to the same disciplined claim boundary.

A framework can become more trustworthy by saying less, checking more, and refusing false certainty. The historical v10.8 line should be read in that spirit: a clearer, more integrated release foundation, not a claim of empirical completion.

The practical path remains: reference calculator, schemas, test vectors, replayable PCC examples, independent reconstruction, inter-rater reliability, and shadow-mode pilots.

Section 17. What the Core Foundation Does and Does Not Include

The core foundation explains the decision architecture, claim boundaries, and public-facing logic of MathGov and RippleLogic. It does not bundle all technical tooling by itself. Schemas, calculators, validators, machine-readable test vectors, evidence-hash registries, and replayable PCC examples belong in companion tooling releases unless explicitly included in the release manifest.

The practical path is staged: v10.8 was the clean core foundation; the current v11.4 line preserves that foundation and adds Term Discrimination, while the tooling line develops replayability; later empirical releases must test inter-rater reliability, reconstructability, and shadow-mode pilots before stronger operational-readiness claims are made.

Final release-readiness interpretation. This primer is release-clean as an informative explanatory companion only. It should not be cited as a normative source of conformance, empirical validation, ProofPack readiness, machine-verifiable ecosystem completeness, or Tier 4 readiness.

Glossary Binding

All technical terms used in this primer (Rights Floor/RF, NCRC, TRC, CSV, RLS, UCI, HOI, PCC, NCAR, HDW, PLSS, SGP, SG_norm, SG_patient_norm, plateau, rights-admissible, admissible, selectable, selected, Refusal, ProjectionPreRegistration, ComputationalClosure, CategoryCollapseSelfTest, RealityGroundingRecord, RealityGroundingStatus, CMIU, UBR, UBE, MNA, Bridge Premises, Stewardship, Execution Stability Diagnostic, PCC.ExecutionStabilityRecord, ESC_STABLE, ESC_DESTABILIZING_BOUNDED, ESC_ENDPOINT_REFUSED, ESC_UNDERDETERMINED, ripple.md, Aligners) are defined canonically in RippleLogic Canon v11.4 Appendix L and SGP v6.4. This primer defers to those definitions and adds none of its own.

Version History

v2.6 (Informative). Synchronized to Core Release 2026.07 and companion artifacts: RippleLogic Canon v10.8 / SGP v5.5 / ripple.md v3.6 / Agent System v10.8 / Aligners Sheet v3.6. Added the Reality Grounding and Lean Cascade Patch, Category Grounding / Primitive Audit explanation, structure-transition-consequence vocabulary, measurement-maturity and deployment-context clarification, HDW capture, PLSS escalation, UCI readiness, kernel posture, biological SGP maturity, and release-boundary cleanup.

v2.1 (Informative). Re-pinned to RippleLogic Canon v10.1 / SGP v4.9 / ripple.md v3.4 / Agent System v10.1 / Aligners v3.4; added transition-admissibility and evidence-humility explanation. No normative claims added; companion-to-canon role unchanged.

v2.0 (Informative). Promoted from v1.5 to v2.0 to reflect added Section 13 (What v10.0 Adds: Reality-Coupled Methodology) and re-pinning to RippleLogic Canon v10.0 / SGP v4.8 / ripple.md v3.0 / Agent System v10.0 / Aligners v3.0. No normative claims added; companion-to-canon role unchanged.

v1.5 (Informative). Added a plain-language Execution Stability section, updated release-line pins to v10.0 / SGP v4.8 / ripple.md v3.0, and clarified that stability is a transition diagnostic subordinate to rights, tail-risk, containment, and legitimate authority.

v1.4 (Informative). Strengthened the reader guide, added explicit layer discipline, softened union language to avoid final-ontology overclaiming, clarified Refusal in plain language, added a simple low-stakes reading example, expanded glossary binding for v9.6.5 hardening terms, and preserved the v9.6.5 / SGP v4.7.2 / ripple.md v2.3.5 companion line.

v1.2.1 (Informative). Final-fix release. Corrected Section 10 subsection numbering from 9.x to 10.x (carry-over from v1.2.0 renumbering). Replaced docx-js auto-numbering with manually-prefixed paragraphs in Section 6 and Section 11 to fix a rendering issue where some viewers reset every list item to “1.” Softened the intelligence-and-isolation passage in Section 4 to avoid unnecessary objections about AI embodiment and absolute scientific claims.

v1.2.0 (Informative). Added Section 4, Union as a Reality Principle. This section unpacks the meaning of UBR for educated general readers, distinguishes union from sameness, names the asymmetric propagation of harm and care, and explicitly rules out three common misreadings: cosmic merger, required opposition (false balance), and collectivist dissolution. Sections 5 through 12 are the v1.0.0 sections renumbered.

v1.0.0 (Informative). Initial Foundations Primer companion to RippleLogic Canon v9.6.5. Established twelve-primitive, eleven-assumption structure with explicit boundary statement, cascade naming, decision-state vocabulary, critical separations, SGP plateau and scalar binding sections, and eleven-step operating sequence including PCC and NCAR.

Appendix A: Position, Precedence, and Release Package

This appendix preserves the primer's informative status, pinning, reader guide, and release-package orientation moved out of the opening pages in v2.6. It does not create normative force and does not alter the Canon.

Section 0. Position and Precedence

This document is an Informative companion to MathGov Core Release 2026.09, governed by RippleLogic Canon v11.4. Its purpose is to make the foundations of MathGov, RippleLogic, and the Sentience Gradient Protocol (SGP) understandable to educated human readers in plain language, without weakening or replacing the normative content of the Canon.

Normative status: This document is Informative. It carries no normative force. Where any sentence in this primer appears to conflict with the canonical specifications, the canonical specifications govern in the order set out in Canon Section 0.1.

Pinning: This primer is pinned to RippleLogic Canon v11.4, SGP v6.4, ripple.md Standard v4.4, RippleLogic Agent System v11.4, and Aligners Sheet v4.4. Any successor release line should re-pin this primer or supersede it explicitly.

Audience: Policy practitioners, governance reviewers, legal staff, public-sector technologists, and educated general readers who need to understand what MathGov claims, what it does not claim, and how its decision architecture works in operational terms.

This primer is explanatory only and does not itself create conformance claims; release identity, claim scope, and normative force remain governed by the pinned canonical artifacts and release manifest.

Reader Guide

If you are new to MathGov, read Sections 1, 2, 4, 7, 8, and 12 first. These sections explain the purpose, boundaries, union logic, cascade, decision states, and governing posture of the framework in plain language.

If you are reviewing the system for technical or institutional use, read Sections 3, 3A, 3B, 5, 9, 10, 11, and 13 next. These sections identify the declared primitives, assumptions, critical separations, SGP interface, and operating sequence.

If a term in this primer appears to conflict with the canon, the canon controls. This primer is an explanatory bridge, not the source of normative force.

How to read the release package

The Canon is the governing specification. SGP is the moral-status and protection protocol. ripple.md is the portable wrapper and assurance contract. The Agent System is the deployment-control architecture. The Aligners Sheet is a worked-run exemplar and training

companion, not a validator. The Methodology Summary explains what changed. The Release Claims and Non-Claims Statement explains what is and is not being claimed publicly.

A simple rule: the core tells you what the system is; the companion files tell you how to understand, test, audit, and eventually validate it.

Section 18. Historical v2.6 Update: Computable Is Not Automatically Admissible

18.1 Plain-language rule

The carried-forward v10.8 line added one important clarity rule: a number can exist without being allowed to decide. A spreadsheet can calculate a score for an option that already failed rights protection. An AI can produce a confident recommendation that lacks authority. A sentence score can be estimated while still being too immature for a strong claim. In those cases, the output is computable but inadmissible.

18.2 Why this matters

This protects the framework from false precision. It prevents dashboards, agents, and reports from laundering weak evidence or failed gates into strong decisions. MathGov remains calculable, but calculation remains subordinate to rights, evidence, containment, lawful authority, and claim boundaries.

18.3 Adaptive reserve

The carried-forward v10.8 line also added an Adaptive Reserve diagnostic. Some systems fail because they are too chaotic; others fail because they are too tightly optimized and have no slack left. A decision can look efficient while making a family, institution, ecosystem, or society brittle. Adaptive Reserve asks whether enough recovery capacity, redundancy, attention, and reversibility remain after the decision.

18.4 v2.6 release delta

The Primer v2.6 explains the Canon v10.8 / SGP v5.5 layer-discipline update in plain language, including Canon Appendix AE. It remains informative only and does not replace the Canon, SGP, ripple.md, Agent System, or Aligners Sheet.

Section 19. Historical v2.6.1 Update: Framework Verdict Is Not Authority Selection

19.1 Plain-language rule

MathGov can say what is blocked, what remains admissible, what is non-decisive, and what requires review. When human, institutional, democratic, expert, emergency, or operator

judgment chooses among admissible non-decisive options, that choice must be labeled as an authority selection, not as a unique mathematical verdict.

19.2 Why this matters

This protects MathGov from false certainty. A dashboard may compute scores and a governance process may choose a path, but the system must not pretend that a non-decisive score created a unique framework-determined answer. Refusal is not failure; it is the disciplined boundary between evidence-backed decision support and overclaim.

19.3 Coverage and projection discipline

Required unknowns must be named. Missing evidence cannot silently become zero. High-stakes runs should state whether scenarios, weights, thresholds, uncertainty assumptions, stakeholder maps, and refusal conditions were declared before evaluation. A changed projection or changed assumption creates a new run boundary.

19.4 v2.6.1 release delta

The Primer v2.6.1 explains the MathGov-native hardening layer: decision-verdict semantics, authority-selection separation, projection-freeze visibility, coverage-status discipline, and falsification-class reasoning. It remains informative only and carries no normative force over the Canon or companion specifications.

Section 20. Historical v2.6 Update Summary: Reality Grounding and Lean Cascade

The v2.6 primer teaches the five-level method: Reality Grounding -> Rights Floor -> TRC -> CSV -> RLS.

Public mantra: Ground -> Protect -> Bound -> Contain -> Score.

UCI/HOI are explained as structural sensors used inside Containment when material and only as special-case tie-break or monitoring tools when RLS is tied, close, or non-decisive.

Section 21. Category Grounding Patch Summary

The Category Grounding patch clarifies that Category Grounding, Primitive Audit, Reference Category Review, Definition Falsification Test, and Category-Reality Correspondence Check belong to one group inside Reality Grounding when material. They are not separate headline gates. They help MathGov avoid optimizing inherited assumptions or building governance layers on top of unexamined definitions.

Public mantra update: Ground the category, ground the surface, protect the floor, bound ruin, contain structural damage, then score only what remains admissible. The short five-level public method remains Ground -> Protect -> Bound -> Contain -> Score.

Section 22. Term Discrimination: The Words Must Hold

MathGov now makes one hidden discipline explicit: before a system governs with a word, it must know what the word means, what it excludes, what separates it from neighboring words, and when the word is too weak for the claim being made.

This is Term Discrimination. It belongs inside Category Grounding, which belongs inside Reality Grounding. It is not a new sixth step. The public sequence remains: Reality Grounding -> Rights Floor -> TRC -> CSV -> RLS.

A material term should answer five questions:

1. What does this term mean in this run?
2. What does it exclude?
3. What discriminator separates it from nearby terms?
4. Where does it sit in the dependency order of the decision?
5. What condition forces narrowing, escalation, monitoring-only use, sensitivity-only use, or refusal?

22.1 Do-not-collapse matrix

Do not collapse	Plain meaning
Reality is not evidence	Evidence is contact with a surface of reality, not the whole territory.
Evidence is not interpretation	Evidence must still be interpreted under uncertainty and values.
A category is not a metric	A metric can only measure what its category correctly defines.
A score is not truth	RLS ranks remaining options; it does not prove reality.
Compliance is not correctness	A system can follow rules that are incomplete or wrong.
Certification is not validity	Certification records evidence; it does not manufacture truth.
Permission is not admissibility	Approval does not mean the option passed MathGov gates.
Monitoring is not control	Watching a system does not mean preventing harm.
Safety is not stability	A safety claim needs bounded conditions; stability needs structural integrity.

Do not collapse	Plain meaning
Stability is not justice	A stable system can still violate rights.
Automation is not autonomy	Automated execution is not accountable self-direction.
Protection is not authority	Moral protection does not grant governance power.
Selected is not executable	A selected option still needs authority and execution controls.
Capability is not authority	Being able to act does not make action legitimate.

Plain-language rule. Bad words create bad measurements. Bad measurements create bad optimization. Bad optimization creates real harm. MathGov therefore grounds the words before it trusts the score.

22.2 Public examples

A school that defines learning only as test-score improvement may optimize the dashboard while weakening real understanding. A company that defines productivity only as output per hour may improve reports while hollowing trust, health, and retention. An AI lab that defines safety as monitoring may watch risks it has not actually controlled. A regulator that treats certification as validity may approve a system whose category, evidence, or reference structure was never strong enough for the claim.

Term Discrimination is the shield against this drift. It does not make MathGov more complicated for its own sake. It makes the framework harder to game, easier to audit, and more honest about what its words can support.

Section 23. v3.0 Update Summary: Term Discrimination and Semantic Stability

The Primer v3.4 explains the Canon v11.4 / SGP v6.4 / ripple.md v4.4 / Agent System v11.4 / Aligners Sheet v4.4 synchronization. The major v11.4 addition is the Complete Ready Release, CSV Upgrade, and Falsification Discipline Patch, carrying forward the Term Discrimination and Semantic Stability Patch. It adds explicit CSV treatment for Containment and Structural Viability while carrying forward the public discipline of naming the discriminator, dependency position, non-collapse risks, and refusal condition for material terms.

Public mantra update: Ground the words, ground the surface, protect the floor, bound ruin, contain structural damage, then score only what remains admissible. The short five-level public method remains Ground -> Protect -> Bound -> Contain -> Score.

Computable, realizable, and ethical are not the same

A modern AI system can generate far more than reality can carry. RippleLogic therefore refuses the shortcut from possibility to selection. A generated option may be computable, fluent, executable by a tool, or attractive in a dashboard and still fail the cascade.

The public rule is: **Computable is not selectable. Generated is not grounded. Executable is not ethical. Realizable is necessary but not sufficient.**

This is why the cascade begins with Reality Grounding and ends with RLS only after Rights Floor, TRC, and CSV have formed the selectable set.

HISTORICAL LINEAGE - NOT CURRENT SPEC. Current active component versions are: Canon v11.4 / SGP v6.4 / ripple.md v4.4 / Agent v11.4 / Primer v3.4 / Aligners Sheet v4.4. Historical passages preserve provenance only and MUST NOT be read as active pins.

Historical lineage note: release-history sections are informative and are not current governing specifications. Current active pins remain Canon v11.4, SGP v6.4, Agent System v11.4, ripple.md v4.4, Primer v3.4, Aligners Sheet v4.4, Cascade Standard v1.0, and CSV Gate Standard v1.2.